

Windows Server Core Infrastructure Administration

Area of Focus Window Server / Active Directory / Network Infrastructure

Project Overview

This project demonstrates hands-on Windows Server 2022 systems administration in a VMware lab environment. The project is designed around common responsibilities found in an IT Systems Administrator role, including server assessment, core infrastructure services, networking, Active Directory, DNS, DHCP, file services, security and patching, PowerShell administration, monitoring, troubleshooting, and technical documentation.

The objective is to build and validate a stable Windows Server environment while documenting each major administrative action, the reason for the change, verification evidence, and any troubleshooting performed.

Key Skills Demonstrated

- Windows Server 2022 administration
 - IPv4 and Windows network configuration
 - Active Directory Domain Services (AD DS)
 - DNS and DHCP administration
 - Windows file server and NTFS/share permissions
 - Windows Firewall and patch-management verification
 - PowerShell administration and basic automation
 - System monitoring and service
 - VMware virtualization
 - Technical documentation and change validation
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Environments & Technologies Used

- VMware Workstation Pro
- Windows Server 2022
- Log N Pacific Server VM
- Windows PowerShell
- Server Manager
- Active Directory Domain Services (AD DS)
- DNS Server

- DHCP Server
- File and Storage Services
- Windows Defender Firewall
- Windows Update
- Event Viewer



Operating Systems

- Windows Server 2022 (Domain Controller)
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Tasks Performed

- Documented the existing Windows Server configuration before making changes
 - Validated server identity, operating system, hardware resources, and installed roles
 - Assessed IPv4, gateway, DNS, DHCP, and network connectivity
 - Configured or validated Active Directory and DNS services
 - Configured or validated DHCP services and address allocation
 - Created and secured a Windows file share using group-based permissions
 - Reviewed Windows Firewall and server update status
 - Used PowerShell for administration, verification, and basic automation
 - Reviewed server health, services, logs, and performance indicators
 - Completed a Tier 2 break/fix troubleshooting scenario
 - Performed final validation and documented project results
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Project Workflow

- Baseline and change assessment
- Server and network configuration
- Active Directory and DNS
- DHCP administration
- File server and permissions
- Security and patch management
- PowerShell administration and automation
- Monitoring and availability checks
- Tier 2 troubleshooting / break-fix
- Final validation and documentation

This workflow mirrors real-world help desk procedures and emphasizes security and accountability.

 Screenshots & Explanations

Existing Server Network Configuration

```
PS C:\Users\Administrator> ipconfig /all

Windows IP Configuration

    Host Name . . . . . : DC01
    Primary Dns Suffix . . . . . : logn.local
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : logn.local

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix . : 
    Description . . . . . : Intel(R) 82574L Gigabit Network Connection
    Physical Address. . . . . : 00-0C-29-7C-76-02
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::4ccd:ae03:fddb:c8f0%6(Preferred)
    IPv4 Address. . . . . : 192.168.4.10(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.4.1
    DHCPv6 IAID . . . . . : 100666409
    DHCPv6 Client DUID. . . . . : 00-01-00-01-32-05-18-5E-00-0C-29-7C-76-02
    DNS Servers . . . . . : ::1
                           127.0.0.1
    NetBIOS over Tcpip. . . . . : Enabled
```

Existing Windows Server Roles

```
PS C:\Users\Administrator> Get-WindowsFeature AD-Domain-Services,DNS,DHCP,File-Services

Display Name                                     Name                               Install State
-----
[X] Active Directory Domain Services             AD-Domain-Services                Installed
[X] DHCP Server                                  DHCP                               Installed
[X] DNS Server                                   DNS                                Installed
[X] File and iSCSI Services                      File-Services                      Installed
```

Static Network Configuration

```
PS C:\Users\Administrator> ipconfig /all
```

```
>> ping
```

```
>> nslookup
```

```
>> Test-NetConnection
```

Windows IP Configuration

```
Host Name . . . . . : DC01
Primary Dns Suffix . . . . . : logn.local
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : logn.local
```

Ethernet adapter Ethernet0:

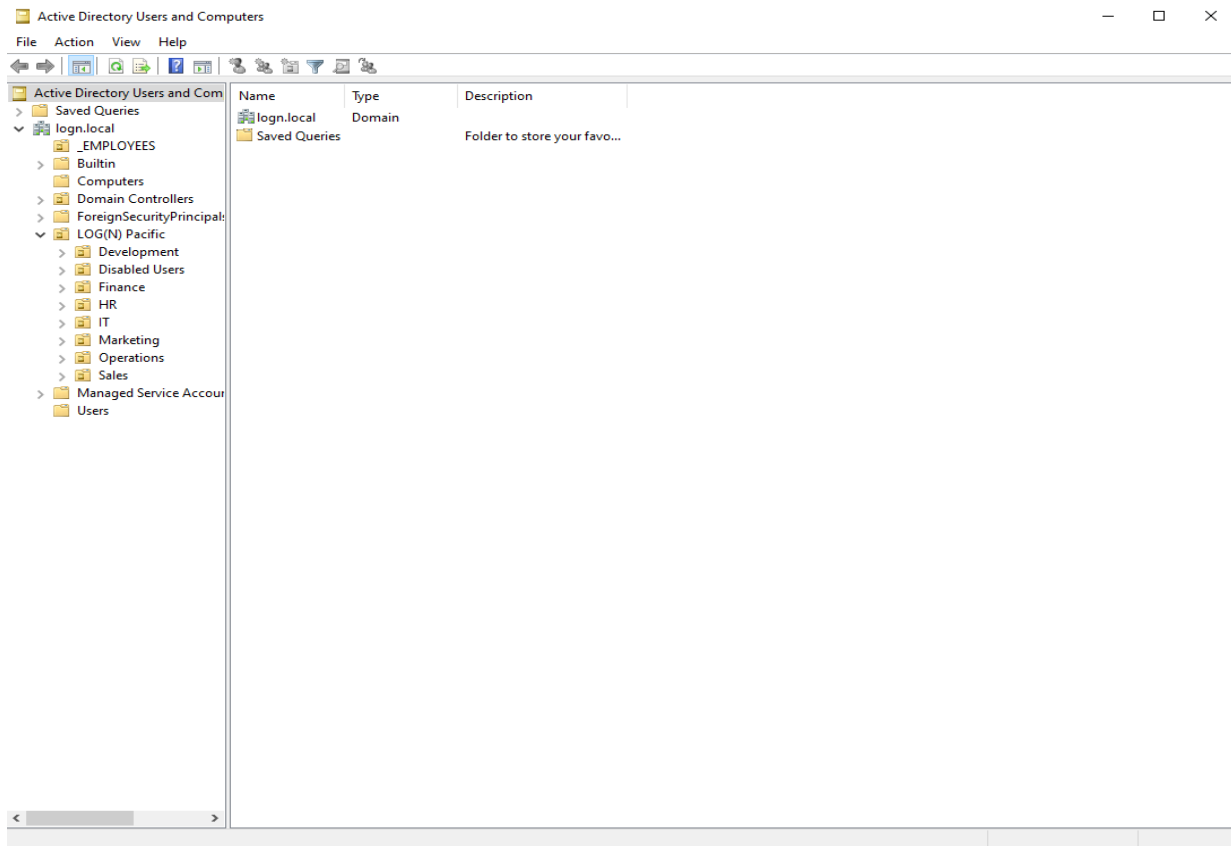
```
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) 82574L Gigabit Network Connection
Physical Address. . . . . : 00-0C-29-7C-76-02
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::4ccd:ae03:fddb:c8f0%6(Preferred)
IPv4 Address. . . . . : 192.168.4.10(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.4.1
DHCPv6 IAID . . . . . : 100666409
DHCPv6 Client DUID. . . . . : 00-01-00-01-32-05-18-5E-00-0C-29-7C-76-02
DNS Servers . . . . . : ::1
                        127.0.0.1
NetBIOS over Tcpip. . . . . : Enabled
```

```
Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
          [-r count] [-s count] [[-j host-list] | [-k host-list]]
          [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
          [-4] [-6] target_name
```

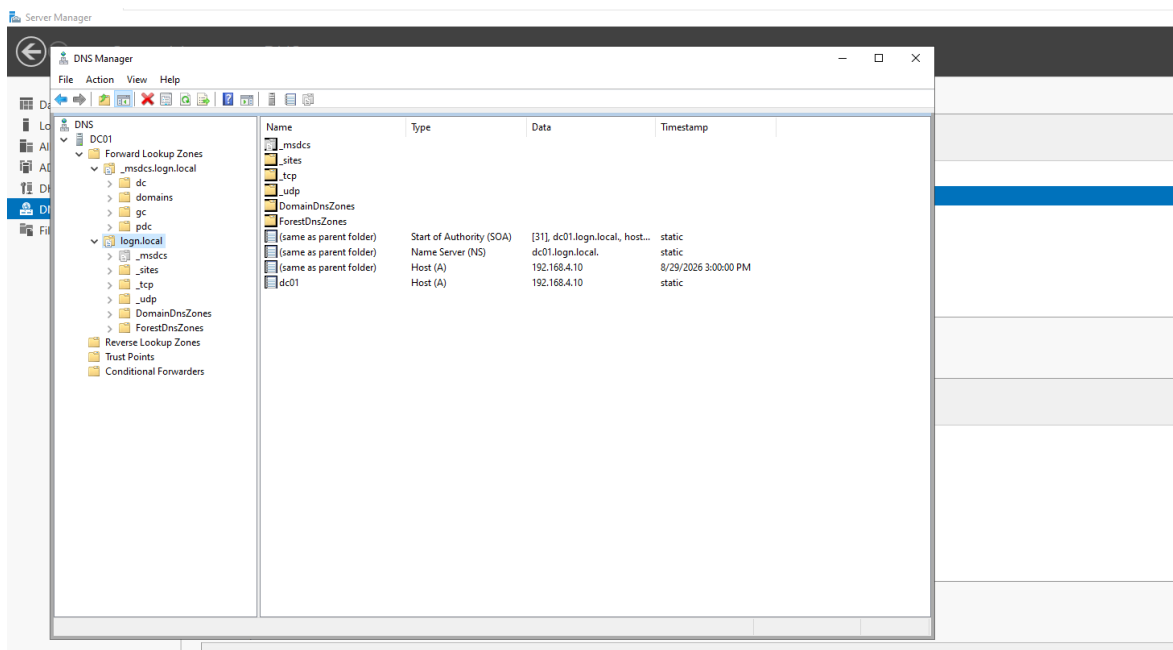
Options:

-t	Ping the specified host until stopped. To see statistics and continue - type Control-Break; To stop - type Control-C.
-a	Resolve addresses to hostnames.
-n count	Number of echo requests to send.
-l size	Send buffer size.
-f	Set Don't Fragment flag in packet (IPv4-only).
-i TTL	Time To Live.
-v TOS	Type Of Service (IPv4-only. This setting has been deprecated and has no effect on the type of service field in the IP Header).
-r count	Record route for count hops (IPv4-only).
-s count	Timestamp for count hops (IPv4-only).
-j host-list	Loose source route along host-list (IPv4-only).
-k host-list	Strict source route along host-list (IPv4-only).
-w timeout	Timeout in milliseconds to wait for each reply.
-R	Use routing header to test reverse route also (IPv6-only). Per RFC 5095 the use of this routing header has been deprecated. Some systems may drop echo requests if this header is used.
-S srcaddr	Source address to use.
-c compartment	Routing compartment identifier.
-p	Ping a Hyper-V Network Virtualization provider address.
-4	Force using IPv4.
-6	Force using IPv6.

Active Directory Users and Computer – OU Structure



DNS Records



Active Directory Users and Computers [DC01.logn.local]

File Action View Help

Active Directory Users and Computers [DC01.logn.local]

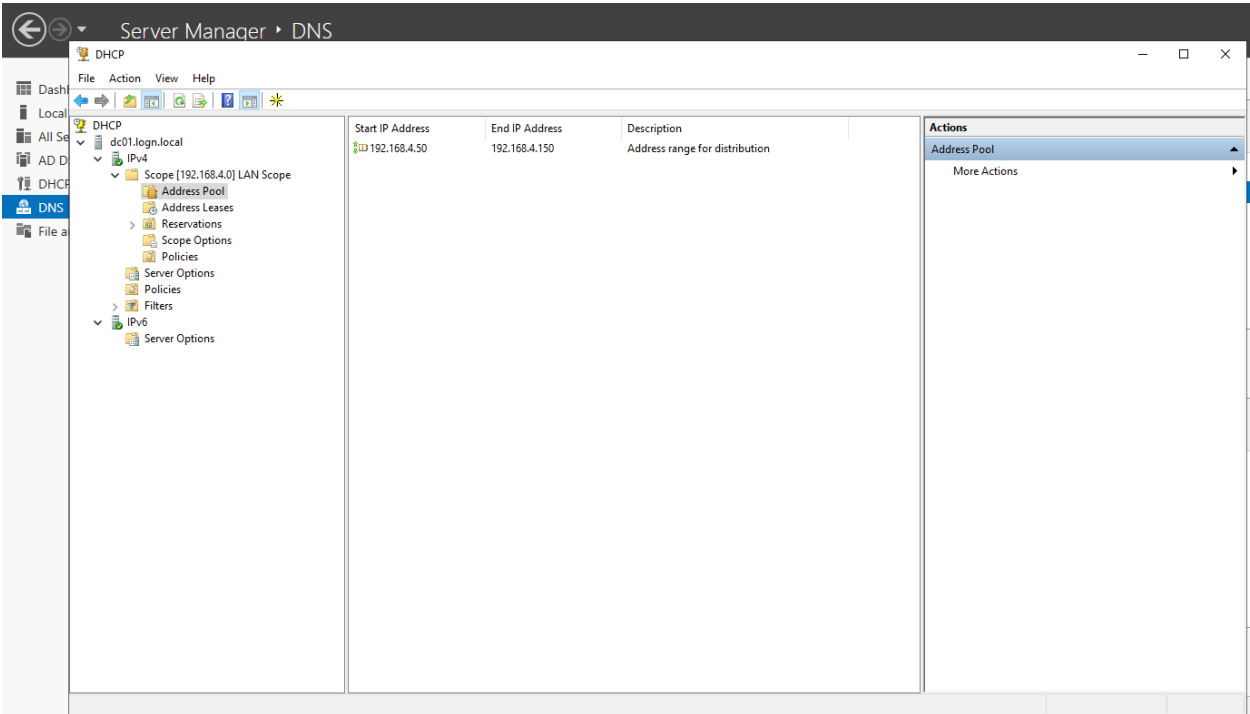
- Active Directory Users and Computers [DC01.logn.local]
 - Saved Queries
 - logn.local
 - EMPLOYEES**
 - Builtin
 - Computers
 - Domain Controllers
 - ForeignSecurityPrincipals
 - LOG(N) Pacific
 - Development
 - Disabled Users
 - Finance
 - HR
 - IT
 - Marketing
 - Operations
 - Sales
 - Managed Service Accounts
 - Users

Name	Type	Description
Bitet Segehoc	User	
Fala Likodi	User	
Hala Larefid	User	
Laseti Kedom	User	
Mifa Fobehag	User	
Miranda Cosgrove	User	
Pabati Tikaj	User	
Steve Jobs	User	
Tony Stark	User	

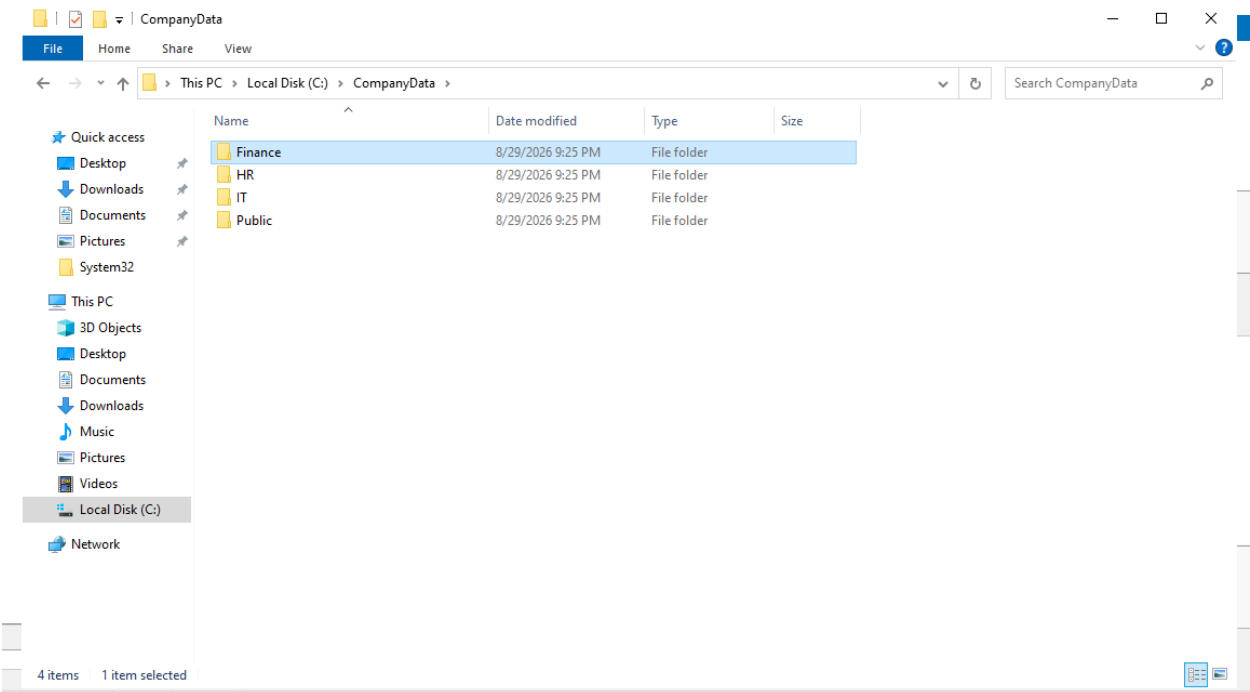
The screenshot shows the 'Active Directory Users and Computers' console for the 'DC01.logn.local' domain. The left pane displays a tree view of the directory structure. The 'logn.local' domain is expanded, showing various built-in and custom containers. The 'LOG(N) Pacific' container is selected and expanded, revealing a list of organizational units (OUs) and security groups. The right pane displays a table of these objects, categorized by type and providing a description.

Name	Type	Description
Development	Organizational Unit	
Disabled Users	Organizational Unit	
Finance	Organizational Unit	
HR	Organizational Unit	
IT	Organizational Unit	
Marketing	Organizational Unit	
Operations	Organizational Unit	
Sales	Organizational Unit	
Development-SG	Security Group - Global	
Finance-SG	Security Group - Global	
HR-SG	Security Group - Global	
IT-Support-SG	Security Group - Global	
Marketing-SG	Security Group - Global	
Operations-SG	Security Group - Global	
Sales-SG	Security Group - Global	

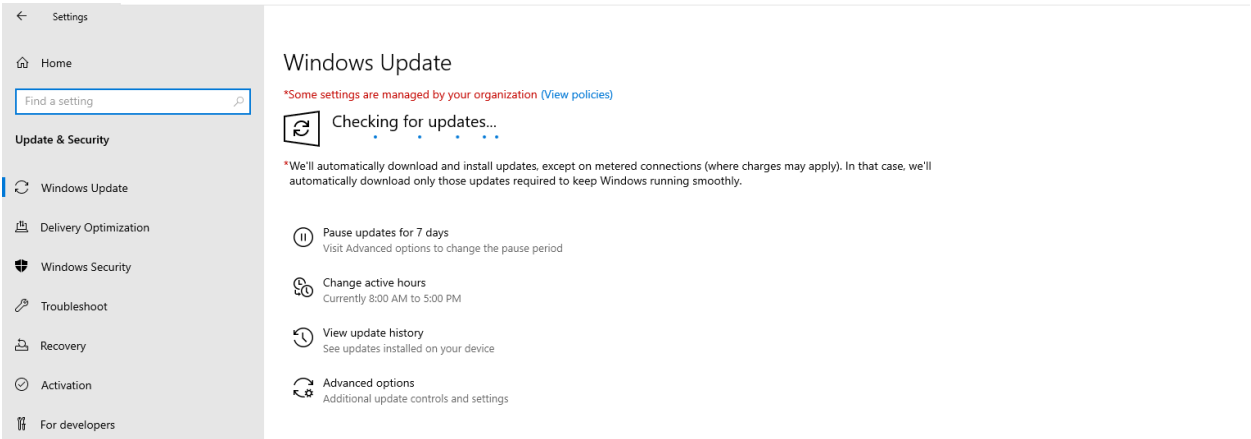
DHCP Scope



Company Data – Folders



Security & Patch Management



PowerShell Server Health Automation

```
Select Administrator: Windows PowerShell
Windows PowerShell
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PS C:\Users\Administrator> Get-Service
>> Get-Volume
>> Get-NetIPConfiguration
>> Get-WindowsFeature

Status  Name                DisplayName
-----
Running ADWS                Active Directory Web Services
Stopped AJRouter          AllJoyn Router Service
Stopped ALG              Application Layer Gateway Service
Stopped AppIDSvc         Application Identity
Stopped Appinfo          Application Information
Stopped AppMgmt          Application Management
Stopped AppReadiness     App Readiness
Stopped AppVClient       Microsoft App-V Client
Running AppXSvc          AppX Deployment Service (AppXSVC)
Stopped AudioEndpointBu... Windows Audio Endpoint Builder
Stopped Audiosrv         Windows Audio
Stopped AxInstSV        ActiveX Installer (AxInstSV)
Running BFE              Base Filtering Engine
Stopped BITS            Background Intelligent Transfer Ser...
Running BrokerInfrastru... Background Tasks Infrastructure Ser...
Stopped bthserv          Bluetooth Support Service
Running camsvc           Capability Access Manager Service
Stopped CaptureService_... CaptureService_3ba766
Running cbdhsvc_3ba766   Clipboard User Service_3ba766
Running CDPSvc           Connected Devices Platform Service
Running CDPUserSvc_3ba766 Connected Devices Platform User Ser...
Stopped CertPropSvc      Certificate Propagation
Stopped ClipSVC          Client License Service (ClipSVC)
Running COMSysApp        COM+ System Application
Stopped ConsentUxUserSv... ConsentUX User Service_3ba766
Running CoreMessagingRe... CoreMessaging
Stopped CredentialEnrol... CredentialEnrollmentManagerUserSvc_...
Running CryptSvc         Cryptographic Services
Stopped CscService       Offline Files
Running DcomLaunch       DCOM Server Process Launcher
Stopped dcsvc            Declared Configuration(DC) service
Stopped defragsvc        Optimize drives
Stopped DeviceAssociati... DeviceAssociationBroker_3ba766
Stopped DeviceAssociati... Device Association Service
Stopped DeviceInstall    Device Install Service
Stopped DevicePickerUse... DevicePicker_3ba766
Stopped DevicesFlowUser... DevicesFlow_3ba766
Stopped DevQueryBroker   DevQuery Background Discovery Broker
Running Dfs              DFS Namespace
Running DFSR             DFS Replication
Running Dhcp             DHCP Client
Running DHCPServer       DHCP Server
Stopped diagnosticshub... Microsoft (R) Diagnostics Hub Stand...
Running DiagTrack        Connected User Experiences and Tele...
Running DispBrokerDesk... Display Policy Service
Stopped DmEnrollmentSvc  Device Management Enrollment Service
Stopped dmwappushservice  Device Management Wireless Applicat...
Running DNS              DNS Server
Running Dnscache         DNS Client
Running DoSvc            Delivery Optimization
```



```

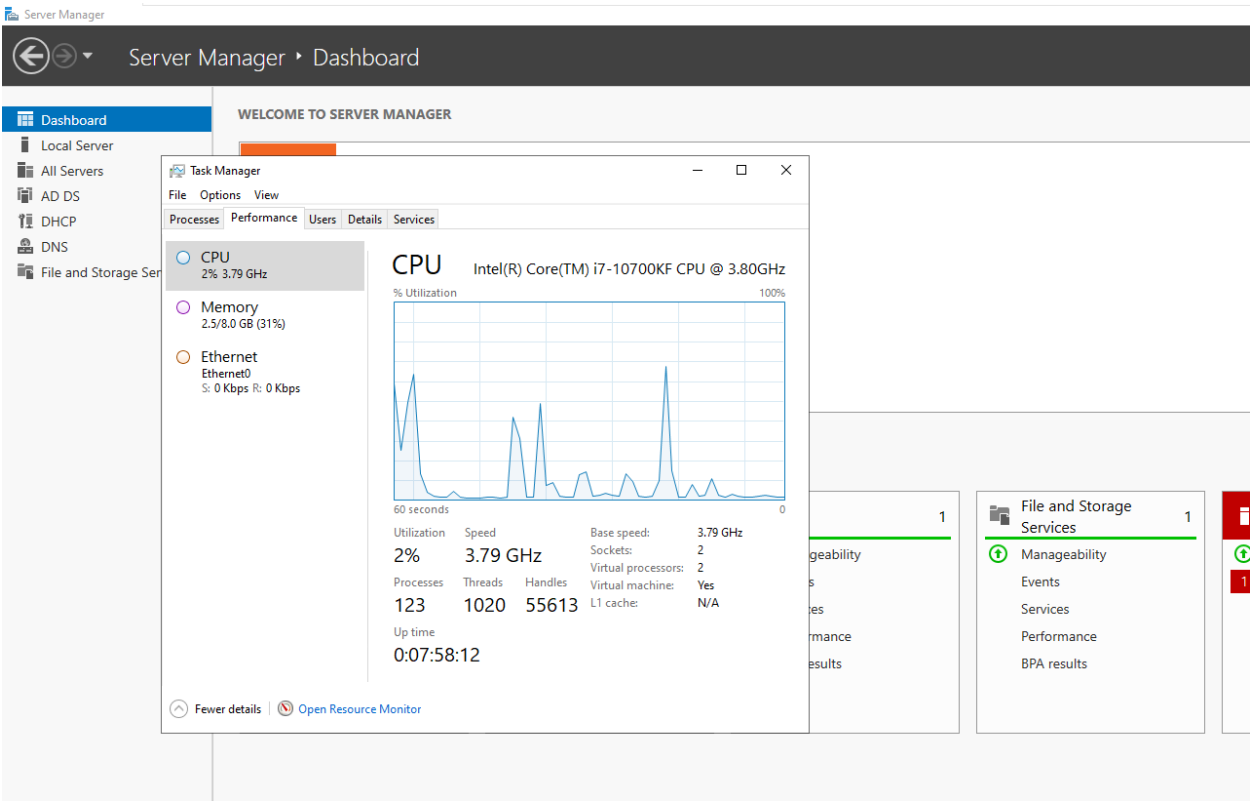
        adapters, and manages both wireless connections and the wireless profiles that contain the
        settings required to configure a wireless client to connect to a wireless network.
Installed      : False
InstallState   : Available
FeatureType    : Feature
Path           : Wireless LAN Service
Depth          : 1
DependsOn      : {}
Parent         :
ServerComponentDescriptor : ServerComponent_Wireless_Networking
SubFeatures    : {}
SystemService  : {}
Notification    : {}
BestPracticesModelId :
EventQuery     :
PostConfigurationNeeded : False
AdditionalInfo  : {MajorVersion, MinorVersion, NumericId, InstallName}

Name           : WoW64-Support
DisplayName     : WoW64 Support
Description     : Includes all of WoW64 to support running 32-bit applications on Server Core installations.
                  This feature is required for full Server installations. Uninstalling WoW64 Support will
                  convert a full Server installation into a Server Core installation.
Installed      : True
InstallState   : Installed
FeatureType    : Feature
Path           : WoW64 Support
Depth          : 1
DependsOn      : {}
Parent         :
ServerComponentDescriptor : ServerComponent_WoW64_Support
SubFeatures    : {}
SystemService  : {}
Notification    : {}
BestPracticesModelId :
EventQuery     :
PostConfigurationNeeded : False
AdditionalInfo  : {MajorVersion, MinorVersion, NumericId, InstallName}

Name           : XPS-Viewer
DisplayName     : XPS Viewer
Description     : The XPS Viewer is used to read, set permissions for, and digitally sign XPS documents.
Installed      : True
InstallState   : Installed
FeatureType    : Feature
Path           : XPS Viewer
Depth          : 1
DependsOn      : {}
Parent         :
ServerComponentDescriptor : ServerComponent_XPS_Viewer
SubFeatures    : {}
SystemService  : {}
Notification    : {}
BestPracticesModelId :
EventQuery     :
PostConfigurationNeeded : False
AdditionalInfo  : {MajorVersion, MinorVersion, NumericId, InstallName}

```

Windows Server Health Monitoring



[Scenario - Tier 2 Infrastructure Troubleshooting] *Finance Shared Folder Access Failure*

A Finance employee reports that they can sign into the domain normally, but they can no longer access the Finance departmental shared folder. Other server functionality appears to be working.

Tier 2 Troubleshooting — Root Cause Identified

```
Administrator: Windows PowerShell
Windows PowerShell
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PS C:\Users\Administrator> Get-SmbShare -Name "Finance"

Name      ScopeName Path              Description
-----
Finance *      C:\CompanyData\Finance

PS C:\Users\Administrator> Get-SmbShareAccess -Name "Finance"

Name      ScopeName AccountName      AccessControlType AccessRight
-----
Finance *      LOGN\Domain Admins Allow             Full
Finance *      LOGN\Finance-SG  Allow             Change

PS C:\Users\Administrator> icacls "C:\CompanyData\Finance"
C:\CompanyData\Finance LOGN\Finance-SG:(OI)(CI)(DENY)(RX)
                        NT AUTHORITY\SYSTEM:(OI)(CI)(F)
                        BUILTIN\Administrators:(OI)(CI)(F)
                        BUILTIN\Users:(OI)(CI)(RX)
                        BUILTIN\Users:(CI)(AD)
                        BUILTIN\Users:(CI)(WD)
                        CREATOR OWNER:(OI)(CI)(IO)(F)

Successfully processed 1 files; Failed processing 0 files
PS C:\Users\Administrator>
```

Tier 2 Troubleshooting — Service Restored

```
PS C:\Users\Administrator> icacls "C:\CompanyData\Finance" /grant "LOGN\Finance-SG:(OI)(CI)M"
processed file: C:\CompanyData\Finance
Successfully processed 1 files; Failed processing 0 files
PS C:\Users\Administrator> icacls "C:\CompanyData\Finance"
C:\CompanyData\Finance LOGN\Finance-SG:(OI)(CI)(M)
                        NT AUTHORITY\SYSTEM:(OI)(CI)(F)
                        BUILTIN\Administrators:(OI)(CI)(F)
                        BUILTIN\Users:(OI)(CI)(RX)
                        BUILTIN\Users:(CI)(AD)
                        BUILTIN\Users:(CI)(WD)
                        CREATOR OWNER:(OI)(CI)(IO)(F)

Successfully processed 1 files; Failed processing 0 files
PS C:\Users\Administrator> Get-SmbShareAccess -Name "Finance"
>> Test-Path "\\DC01\Finance"

Name      ScopeName AccountName      AccessControlType AccessRight
-----
Finance *      LOGN\Domain Admins Allow             Full
Finance *      LOGN\Finance-SG  Allow             Change
True

PS C:\Users\Administrator>
```

Shows the permissions have been restored for the finance department.

Outcome

Successfully configured, administered, secured, monitored, and validated a Windows Server 2022 domain environment representing a small business infrastructure. The project included Active Directory and DNS administration, DHCP services, departmental file storage, security group management, NTFS and SMB permissions, patch and firewall assessment, PowerShell administration, and server performance monitoring.

A simulated Tier 2 incident involving loss of access to the Finance departmental share was also investigated. Through systematic troubleshooting, the issue was isolated to an incorrect NTFS Deny permission assigned to the Finance security group. The permission configuration was corrected, appropriate access was restored, and the environment was successfully validated afterward.

The completed environment demonstrated that the domain controller, DNS, DHCP, Active Directory, SMB file services, security permissions, and core Windows Server services were functioning correctly at the conclusion of the project.

What This Project Demonstrates to Employers

This project demonstrates practical experience performing common responsibilities expected of a Junior Systems Administrator / Systems Administrator, including:

- Windows Server Administration — Configuring and maintaining core Windows Server infrastructure and validating server health and service availability.
- Active Directory Administration — Working with organizational units, users, security groups, and group-based access to company resources.
- DNS & DHCP Administration — Managing essential network services responsible for name resolution and client IP configuration.
- File Server Administration — Creating centralized departmental storage and configuring SMB shares for network access.
- NTFS & Share Permissions — Applying group-based access controls and understanding the interaction between SMB share permissions and NTFS permissions.
- Security Administration — Reviewing Windows Update status, firewall configuration, and access controls as part of routine server security management.

- PowerShell Administration — Using PowerShell to inspect, configure, validate, and troubleshoot Windows Server components.
- Performance Monitoring — Reviewing CPU, memory, disk, network utilization, and other server health indicators.
- Tier 2 Troubleshooting — Investigating a reported departmental file-access incident by eliminating possible causes, identifying the root cause, implementing a corrective action, and validating service restoration.
- Infrastructure Validation — Performing post-configuration checks across Active Directory, DNS, DHCP, SMB, and Windows services to confirm that the environment remained operational.
- Technical Documentation — Documenting configurations, administrative procedures, troubleshooting evidence, screenshots, and final outcomes in a structured project format.